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Learning the Shrinkage Intensity: A Data-Driven Approach for Risk-Optimized Portfolios

Gianluca De Nard [1,2,3] **and Damjan Kostovic** [4]

[1]Liechtenstein Business School, University of Liechtenstein, Vaduz, Liechtenstein
[2]Department of Economics, University of Zurich, Zurich, Switzerland
[3]Zürcher Kantonalbank, Zurich, Switzerland
[4]OLZ AG, Zurich, Switzerland

Address correspondence to Gianluca De Nard, Department of Economics, University of Zurich, Zürichbergstrasse 14, 8032 Zürich, Switzerland, or e-mail: gianluca.denard@econ.uzh.ch.

Abstract

We introduce a new type of shrinkage estimator that is not based on asymptotic optimality, but instead learns a state-dependent shrinkage policy via supervised learning in a contextual bandit setup. The proposed estimator applies to both linear and nonlinear shrinkage and shows improved performance compared to classical shrinkage estimators. Our results demonstrate that our estimator identifies a downward bias in classical shrinkage intensity estimates derived under the i.i.d. assumption and automatically corrects for it in response to prevailing market conditions. Additionally, our data-driven approach enables more efficient implementation of risk-optimized portfolios and is well-suited for real-world investment applications, including portfolios with practical optimization constraints.

Keywords: covariance matrix estimation, linear and nonlinear shrinkage, policy learning, portfolio management, reinforcement learning, risk optimization

JEL classifications: C13, C58, G11

Financial applications such as portfolio optimization, asset pricing, and risk management rely heavily on the covariance matrix of asset returns. Markowitz (1952) recognizes it as a central risk measure in his seminal work, which has since sparked extensive research on estimating both expected returns and the covariance matrix. While much focus has been placed on return or variance estimation, our emphasis lies on estimating the covariance matrix.

The most common estimator, the sample covariance matrix, often performs poorly out-of-sample, especially in high-dimensional settings. When the number of assets N approaches or exceeds the number of time observations T, the sample covariance matrix becomes ill-conditioned or even non-invertible. As the concentration ratio $c := N/T$ increases, estimation errors grow and portfolio optimization becomes unreliable, leading to poor performance.

Jobson and Korkie (1980) and Michaud (1989) highlight that the sample covariance matrix suffers from large estimation errors, often amplifying extreme and unreliable

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coefficients. Ledoit and Wolf (2017) reframe this issue as a degrees-of-freedom problem, where estimating $\mathcal{O}(N^2)$ parameters from limited data is infeasible. Their work and that of others introduced linear and nonlinear shrinkage methods to reduce dimensionality and improve estimation; for an overview see Ledoit and Wolf (2022a).

Linear shrinkage combines the sample covariance matrix with a structured target, reducing extreme values and improving estimation accuracy. Empirical results show it enhances information ratios, diversification, and portfolio stability. In constrained settings, shrinkage can be interpreted as an implied covariance matrix (Roncalli 2011), and is widely used by practitioners (Jagannathan and Ma 2003).

Nonlinear shrinkage, proposed by Ledoit and Wolf (2012), shrinks individual eigenvalues without a fixed target and is particularly effective in high-dimensional contexts. Both linear and nonlinear shrinkage estimators remain invertible for $c > 1$, providing feasible portfolio solutions. However, their derivations assume i.i.d. returns. Although extensions exist (e.g., Sancetta 2008), they can introduce high bias in small samples (Bartz and Müller 2014).

The main contribution of this study is to introduce a data-driven approach to compute the shrinkage intensity without making assumptions on the data generating process, using a supervised policy learning approach. The idea of improving shrinkage estimation of the covariance matrix with reinforcement learning (RL) has been introduced by Matera and Matera (2023) and Lu, Ndiaye, and Simaan (2024). Lu, Ndiaye, and Simaan (2024) show how RL based on alternative data and textual analysis can improve the optimal linear shrinkage policy for correlation matrices. Their focus is on so-called text-based networks (TBN) introduced by Hoberg and Phillips (2016) to determine the correlation matrix shrinkage target and to use deep RL to compute the shrinkage intensity that maximizes the expected out-of-sample utility of an investor in line with the theoretical work by Bodnar, Okhrin, and Parolya (2023). Despite some promising results in their empirical analysis, their approach has some (practical) limitations. First, the TBN based shrinkage target depends on the Hoberg-Phillips Data Library and is therefore only applicable to stocks included in the library and not implementable for real-life trading. Second, it is a shrinkage approach for the correlation matrix, ignoring the large estimation errors coming from the variances. Third, it is limited to linear shrinkage. Fourth, the portfolios are rebalanced, respectively the covariance/correlation matrix is re-estimated, on a yearly basis, which is too infrequent regarding the dynamics of the market and of the optimal shrinkage intensity.[1] Finally, the data-driven shrinkage intensity computation can be applied to all shrinkage targets and is not just restricted to TBN.

A related study of Matera and Matera (2023) uses Fama-French industry portfolio return data to derive the shrinkage intensity for covariance matrix estimation using RL. For one of their models, they report improved out-of-sample Sharpe ratio numbers. However, the RL architecture and empirical analysis of Matera and Matera (2023) is insufficient to evaluate the goodness of a covariance matrix. The review and guide to covariance matrix estimation of Ledoit and Wolf (2022a), as well as the references therein, show that for financial applications, the out-of-sample standard deviation of the global minimum variance (GMV) portfolio should be computed to assess the covariance matrix estimation quality. Estimating the GMV portfolio is a "clean" problem in terms of evaluating the quality of a covariance matrix estimator since it abstracts from having to estimate the vector of expected returns at the same time. Therefore, the approach of Matera and Matera (2023) to implement an RL agent with maximum Sharpe ratio utility function is misleading for covariance matrix estimation. Even though their research design is not suitable to assess the power of RL to improve the shrinkage intensity computation for covariance matrix

[1] The shrinkage intensity can spike during periods of financial turmoil where more shrinkage is needed; for example, see Figure A1 of De Nard (2022).

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estimation compared to Ledoit and Wolf (2004b), it is an interesting alternative to Cong et al. (2021) who use RL to directly derive portfolio weights with maximal Sharpe ratio without estimating the expected returns and the covariance matrix. Additionally, the study does not consider daily stock returns but monthly Fama-French industry portfolio returns instead. This results in a practically unimplementable portfolio and low-frequency analysis with a very short out-of-sample period of 3 years, that is, 36 observations, which is far from the literature standard and it is dangerous to claim (statistically significant) outperformance. Finally, the methodology of Matera and Matera (2023) is only applicable to linear shrinkage and has been analyzed only for the most basic (scalar multiple of the) identity matrix shrinkage target, neglecting the off-diagonal entries of the covariance matrix (see De Nard 2022).

As mentioned above, the main contribution of this study is to introduce a data-driven approach to compute the shrinkage intensity without making assumptions on the data generating process. We propose a new type of shrinkage estimator for both linear and nonlinear shrinkage that is not based on asymptotic optimality, but instead uses a supervised policy learning approach to shrink the sample eigenvalues. More specifically, we learn a mapping from market state variables to shrinkage actions. Because state transitions are typically action-independent in investment problems (as in ours, as well as in Matera and Matera 2023; Lu, Ndiaye, and Simaan 2024), the sequential Markov decision process (MDP) simplifies to a one-step contextual bandit. Additionally, we can quickly evaluate the out-of-sample portfolio performance for a large grid of shrinkage intensities. This leads to a simpler and more transparent solution: (offline) supervised policy learning with full-information labels. The new shrinkage estimator delivers improved estimation accuracy, especially in high-dimensional settings where the sample covariance matrix is ill-conditioned or even non-invertible.

Our proposed shrinkage estimator requires only stock return data, but can be easily extended with factor or alternative data signals. We show that the estimator learns that the shrinkage intensity(ies) derived under the i.i.d. assumption are biased downward and automatically corrects for this bias by increasing the shrinkage intensity(ies) in a manner that depends on the market environment. The learned and adjusted shrinkage intensity(ies) remain stable, intuitive, and correlated with the classical i.i.d. approach.

To assess the quality of our covariance matrix estimators, we run an extensive empirical study in a high-dimensional setting focusing on the out-of-sample standard deviation of GMV portfolios, where we optimize the policy for a minimum-variance utility function to learn the shrinkage intensity(ies). Finally, our covariance matrix estimators deliver a more efficient implementation of risk-optimized portfolios and should be used for real-life investments. However, our goal is to improve the estimation of covariance matrices in general, and our methodology is therefore not restricted to portfolio selection or financial applications in general.

A further contribution of this study is the application of a policy learning framework to constrained portfolio optimization, recognizing its practical relevance and its flexibility to adapt to specific investor preferences. Unlike classical shrinkage methods, which are independent of the optimization step, our optimized policy dynamically adjusts shrinkage based on the specific constraints, risk-return objectives, and, hence, the investor's utility function. Most classical shrinkage methods tend to shrink too aggressively in constrained optimization problems, as they do not account for the implicit shrinkage impact of constraints, potentially leading to suboptimal portfolio allocations. This adaptability allows our estimator to incorporate the interaction between shrinkage and portfolio performance in a forward-looking manner, making it particularly well-suited for real-world investment settings where constraints play a crucial role.

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The rest of the study is organized as follows. Section 1 provides a selective overview of the shrinkage estimation and policy learning literature and presents our methodology on how to combine them. Section 2 contains the backtest results and empirical analysis. Section 3 concludes. Appendices contain additional results and auxiliary materials.

1 Methodology

1.1 Shrinkage Estimation

As the population covariance matrix Σ is not directly observable, it needs to be estimated using historical data. The most straightforward method to estimate Σ is simply to use the sample covariance matrix S:

$$S := \frac{1}{T-1}\tilde{R}'\tilde{R} \ , \tag{1.1}$$

where $\tilde{R}$ denotes the $T \times N$ demeaned asset return matrix, N being the number of assets, and T the number of time periods (observations).

The sample covariance matrix is an intuitive estimator, easy to compute, and unbiased. However, we show that it is unsuitable for portfolio optimization, aligning with and extending the conclusions of Ledoit and Wolf (2022a).

One potential solution to mitigate the impact of estimation error is to impose structure on the covariance matrix, thereby reducing the large proportion of random noise in the sample moments. This approach leads to a more parsimonious parametrization of the covariance matrix, which facilitates the efficient extraction of systematic information from historical correlations. In turn, this reduces estimation error and enhances forecasting accuracy. A significant body of research has explored this approach (e.g., Sharpe 1963; Elton and Gruber 1973; Jobson and Korkie 1980). In this study, we focus specifically on the statistical shrinkage technique introduced by Stein (1956). Subsequent work by Ledoit and Wolf (2004a, 2017), Engle, Ledoit, and Wolf (2019), De Nard et al. (2022), and De Nard, Ledoit, and Wolf (2025) has proposed various linear and nonlinear shrinkage estimators for the covariance matrix and extensions based on factor models (e.g., De Nard, Ledoit, and Wolf 2021; De Nard, Engle, and Kelly 2024; Fan et al. 2024) to this end. To readers not already familiar with these shrinkage estimators, we refer to the overview paper of Ledoit and Wolf (2022a).

1.1.1 Linear shrinkage

As discussed above, linear shrinkage combines a highly structured estimator F and the sample covariance matrix S. Ledoit and Wolf (2003) suggest a convex linear combination of the form:

$$\tilde{\Sigma}_{\mathrm{L}} := \delta F + (1-\delta)S \ , \tag{1.2}$$

where δ is the shrinkage intensity, or shrinkage constant, strictly between 0 and 1, and F denotes the shrinkage target. The sample covariance matrix is therefore shrunk toward the structured estimator with an intensity of δ. Ledoit and Wolf (2003) define this shrinkage constant as the weight that is given to the structured measure and that there should be only one optimal shrinkage constant that minimizes the expected distance between the shrinkage estimator $\tilde{\Sigma}_{\mathrm{L}}$ and the true covariance matrix Σ. Moreover, they propose a loss function $\mathcal{L}_F$ based on the Frobenius norm, a quadratic measure of distance between the true and estimated covariance matrix, that does not depend on the inverse of the covariance matrix (Frost and Savarino 1986) and therefore also works for $N > T$. The optimal shrinkage

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intensity is therefore the constant that minimizes the expected quadratic loss function of the form:

$$\mathbb{E}[\mathcal{L}_F(\delta)] := \mathbb{E}[\|\tilde{\Sigma}_{\text{L}} - \Sigma\|_F^2] = \mathbb{E}[\|\delta F + (1-\delta)S - \Sigma\|_F^2] \ , \tag{1.3}$$

where $\|\cdot\|_F^2$ is the (squared) Frobenius norm: $\|A\|_F^2 := \langle A, A\rangle := \text{tr}(AA') = \sum_{i=1}^N \sum_{j=1}^N a_{ij}^2$ for a squared matrix A with entries $(a_{ij})_{i,j=1,\ldots,N}$.

Under the i.i.d. assumption, the literature derives the optimal shrinkage intensity (and target), and introduces the optimal feasible and consistent linear shrinkage estimator as

$$\widehat{\Sigma}_{\text{L}} := \widehat{\delta} F + (1-\widehat{\delta})S \ , \tag{1.4}$$

$$\widehat{\delta} := \min\left\{\max\left\{\frac{\widehat{\pi} - \widehat{\rho}}{\widehat{\kappa} \cdot T}, 0\right\}, 1\right\}, \tag{1.5}$$

where $\widehat{\pi} := \sum_{i=1}^N \sum_{j=1}^N \frac{1}{T}\sum_{t=1}^T ((r_{it} - \bar{r}_{i\cdot})(r_{jt} - \bar{r}_{j\cdot}) - s_{ij})^2$ measures the estimation uncertainty in the sample covariance matrix, $\widehat{\rho}$ measures the ("combined") covariance between the shrinkage target and the sample covariance matrix,[2] and $\widehat{\kappa} := \sum_{i=1}^N \sum_{i=1}^N (f_{ij} - s_{ij})^2$ measures how close the (population version of the) shrinkage target is to the population covariance matrix; see Ledoit and Wolf (2003, 2004b, 2022a) and De Nard (2022).

A key advantage of $\widehat{\Sigma}_{\text{L}}$ is that it remains positive definite and thus invertible, even when $N > T$, unlike the sample covariance matrix, which becomes rank-deficient and not invertible in such cases. This is due to $\widehat{\Sigma}_{\text{L}}$ being a convex linear combination of two matrices: F, which should be positive definite, and S, which is positive semi-definite.

Therefore, the solution (Equations 1.4 and 1.5) of problem (Equation 1.3) depends on the choice of the shrinkage target. The question arises: what constitutes an effective shrinkage target? Ideally, it should approximate the true covariance matrix as closely as possible while minimizing the number of parameters. This presents a delicate "balancing act" between accuracy and parsimony. A good shrinkage target leverages application-specific knowledge, allowing for customization that enhances its relevance and effectiveness in the given context. For an overview of attractive (custom-tailored) shrinkage target see Ledoit and Wolf (2022a). In this study, we focus on the original proposition of Ledoit and Wolf (2004a) and its generalization of De Nard (2022).

The simplest shrinkage target is to assume that the covariance matrix is a scalar multiple of the identity matrix $F_{\text{I}} = \widehat{\eta} I$, where I is the $N \times N$ identity matrix and $\widehat{\eta}$ is the estimated scalar multiple. Note that η depends on the unobservable parameter Σ, which can be consistently estimated by the sample covariance matrix: $\widehat{\eta} = \frac{\text{tr}(SI)}{N} = \frac{1}{N}\sum_{i=1}^N s_i^2 = \bar{s}^2$. Consequently,

$$F_{\text{I}} := \bar{s}^2 I \ . \tag{1.6}$$

Therefore, the linear shrinkage to the identity matrix approach, shrinks all the sample variances toward their grand mean with the estimated (constant) intensity $\widehat{\delta}^{\text{I}}$. Alternatively, one can reformulate the problem in the eigenvalue space, where all the sample eigenvalues $\lambda_1, \ldots, \lambda_N$ are shrunk toward their grand mean $\bar{\lambda} := \frac{1}{N}\sum_{i=1}^N \lambda_i$ with the same intensity $\widehat{\delta}^{\text{I}}$:

[2] $\widehat{\rho}$ depends on the choice of shrinkage target and thus requires a case-by-case analysis. For the exact definition of $\widehat{\rho}$, we therefore reference to the corresponding original paper. Note that the authors show that the impact of $\widehat{\rho}$ on $\widehat{\delta}$ is usually small and can be neglected.

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$$\lambda_i^{\mathrm{I}} := \widehat{\delta}^{\mathrm{I}}\bar{\lambda} + (1-\widehat{\delta}^{\mathrm{I}})\lambda_i \ , \tag{1.7}$$

while the sample eigenvectors remain unchanged.[3]

A scalar multiple of the identity matrix is a highly structured estimator that generally exhibits lower estimation error compared to the sample covariance matrix. However, it is limited in that it only considers the diagonal elements of the covariance matrix, effectively disregarding the off-diagonal covariance coefficients by setting them to zero. Given that asset returns typically exhibit positive linear dependence, it is crucial to choose a shrinkage target that captures this relationship. In this regard, De Nard (2022) introduces a generalization of the identity matrix shrinkage estimator, which incorporates knowledge of the non-zero covariances in asset returns. This is achieved by introducing an additional (off-diagonal) parameter into the shrinkage target, allowing for a more accurate reflection of the covariance matrix structure:

$$F_{\mathrm{CVC}} := \bar{s}^2 I + \bar{s}_{ij} J \ , \tag{1.8}$$

where $J := \mathbb{1}\mathbb{1}' - I$ is the off-diagonal matrix, $\mathbb{1}$ denotes a conformable vector of ones, and $\bar{s}_{ij} := (N(N-1))^{-1}\sum_{i=1}^{N}\sum_{j=1,j\neq i}^{N} s_{ij}$. Thus, De Nard (2022) proposes not only shrinking the diagonal elements of the sample covariance matrix toward a constant mean variance level, but also shrinking the off-diagonal elements toward a constant mean covariance level. Note that the literature presents even more customized shrinkage targets; for an overview see Ledoit and Wolf (2022a).

1.1.2 Nonlinear shrinkage

The optimal linear combination in Equation (1.4) can be intuitively interpreted as moving each entry of the sample covariance matrix toward the shrinkage target with a common intensity. A natural extension of this approach is to allow for different intensities for different entries of the covariance matrix, or alternatively, shrinking the sample eigenvalues to their grand mean with *individual intensities* while keeping the sample eigenvectors; see Equation (1.7). This latter approach, called nonlinear shrinkage, was introduced and continuously improved by Ledoit and Wolf (2012, 2017, 2020, 2022b).

Nonlinear shrinkage allows sample eigenvalues to be individually moved up or down by an individual amount. Note that this means that the individual shrinkage intensities can be negative, that is, move a sample eigenvalue away from the shrinkage target! Clearly, this more general approach will outperform the use of a common shrinkage intensity, provided that the distinct intensities are selected appropriately. Furthermore, this method will yield a positive-definite estimator, as long as all the transformed eigenvalues remain positive.

The optimization problem for nonlinear shrinkage is as follows:

$$\min_{\tilde{\Lambda}} \|\tilde{\Sigma}_{\mathrm{NL}} - \Sigma\|_F^2 \ , \tag{1.9}$$

where $\tilde{\Sigma}_{\mathrm{NL}} := U\tilde{\Lambda}U'$ is the nonlinear shrinkage estimator, $U := [u_1 \ldots, u_N]$ denotes an orthogonal matrix whose columns are the sample eigenvectors u_i, and $\tilde{\Lambda} := \mathsf{Diag}(\tilde{\lambda}_1, \ldots, \tilde{\lambda}_N)$ denotes the diagonal matrix of the N shrunk eigenvalues.

For the solution of problem (Equation 1.9), $\widehat{\Sigma}_{\mathrm{NL}} := U\Lambda_{\mathrm{NL}}U'$, and thus on the actual estimation of the optimal nonlinear shrinkage eigenvalues, $\Lambda_{\mathrm{NL}} := \mathsf{Diag}(\lambda_1^{\mathrm{NL}}, \ldots, \lambda_N^{\mathrm{NL}})$, we refer to the overview study of Ledoit and Wolf (2022a) and the references therein.

[3] Therefore, $\widehat{\Sigma}_{\mathrm{I}} = \widehat{\delta}^{\mathrm{I}}\bar{s}^2 I + (1-\widehat{\delta}^{\mathrm{I}})S$ is equivalent to $\widehat{\Sigma}_{\mathrm{I}} = U\Lambda_{\mathrm{I}}U'$, where $U := [u_1 \ldots, u_N]$ denotes an orthogonal matrix whose columns are the sample eigenvectors u_i, and $\Lambda_{\mathrm{I}} := \mathsf{Diag}(\lambda_1^{\mathrm{I}}, \ldots, \lambda_N^{\mathrm{I}})$ denotes the diagonal matrix of the N linearly shrunk eigenvalues.

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1.2 Supervised Policy Learning

In a standard RL framework, an agent interacts sequentially with an environment (in our case, a financial market) described by a MDP. Formally, an MDP is characterized by a tuple $\mathcal{M} := \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \zeta \rangle$, where $\mathcal{S}$ denotes the set of states (in our case, observable stock prices and other factors), $\mathcal{A}$ the set of actions (in our case, the shrinkage intensity), $\mathcal{P}$ the state transition probability matrix (in our case, independent of actions), $\mathcal{R}$ the reward function (in our case, the negative portfolio standard deviation), and ζ the discount factor (in our case, equal to one for a one-period model).

In our application, the MDP structure simplifies considerably. Specifically, the environment's state transitions are independent of the actions taken, that is, the actions chosen do not influence the underlying (market) state evolution. Formally, this can be described as an MDP with state-transition probabilities independent of actions:

$$P(s_{t+1}|s_t, a_t) = P(s_{t+1}|s_t), \quad \forall \quad a_t \in \mathcal{A} \ . \tag{1.10}$$

Such a setting naturally reduces the sequential decision-making problem to a series of independent *contextual bandit* problems (see, e.g., Lattimore and Szepesvári 2020). In other words, each state can be optimized individually without regard to future states, and the problem reduces to finding the optimal action for each state separately. Similar approaches have been applied to portfolio selection problems where sequential decision-making simplifies to context-dependent optimization, demonstrating the suitability of contextual bandit frameworks for certain classes of financial problems (see, e.g., Shen et al. 2015).

Additionally, we can "cheaply" evaluate realized portfolio risk (ex post portfolio volatility) for a finite action grid using only information available at t. This yields a full-information label (an oracle action) for that state and transforms shrinkage intensity selection into an offline supervised policy learning problem with full-information labels, also known as oracle imitation or behavioral cloning (see, e.g., Ross and Bagnell 2010; Levine et al. 2020). Our objective is to predict the action a_t that minimizes observed portfolio standard deviation $\widehat{\sigma}(s_t, a_t)$ for a given state s_t[4]:

$$y^*(s_t) = \arg\min_{a_t \in \mathcal{A}} \widehat{\sigma}(s_t, a_t) \ . \tag{1.11}$$

We implement a regression-based policy: for each state s_t, a regression model outputs a continuous action prediction, which we map to the discrete action set $\mathcal{A} = \{0, 1, \ldots, 100\}$ by rounding to the nearest integer. Conceptually, this differs from classical RL. Because the market dynamics in our setting do not depend on the chosen action, there is no multi-step credit assignment problem: the optimal decision at time t can be treated myopically. Moreover, for each observed state s_t we can directly evaluate realized portfolio risk across the finite action grid and recover an oracle action $y^*(s_t)$ that minimizes ex post volatility. This provides full-information labels for the optimal action and removes the usual exploration–exploitation trade-off associated with trial-and-error policy improvement. As a result, we do not estimate action-value functions or learn a transition model. Instead, we train the policy in a purely supervised fashion to imitate the oracle action in each state.

The optimal policy is given by

$$\pi^*(s) = \arg\min_{\pi \in \Pi} \mathbb{E}_s[(\pi(s) - y^*(s))^2] \ , \tag{1.12}$$

where Π denotes the space of admissible policies.

[4] The observed portfolio (sample) standard deviation from an action a_t (shrinkage intensity) at state s_t is $\widehat{\sigma}(s_t, a_t) = \sqrt{\frac{1}{\tilde{T}-1} \sum_{\tilde{t}=t+1}^{\tilde{t}+\tilde{T}-1} \left(r_{p,\tilde{t}}(s_t, a_t) - \bar{r}_p \right)^2}$, where $r_{p,\tilde{t}}$ is the observed portfolio return at time $\tilde{t}$, $\tilde{T}$ is the portfolio holding period length, and $\bar{r}_p$ is the mean portfolio return over the holding period.

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In our case, the optimal policy is approximated by empirical risk minimization:

$$\widehat{\pi}(s) = \arg\min_{f \in \mathcal{F}} \sum_{t=1}^{T} \left(f(s_t) - y^*(s_t) \right)^2 , \tag{1.13}$$

where $f(s_t)$ denotes the predicted continuous action for state s_t, $y^*(s_t)$ is the historically observed optimal discrete action that minimizes portfolio standard deviation (oracle action), and $\mathcal{F}$ is the regression model function class, which we use to approximate policies from the broader policy space Π. Our regression-based policy $\widehat{\pi}(s)$ provides an explicit approximation to the optimal policy $\pi^*(s)$ by minimizing squared deviations from historically optimal actions $y^*(s_t)$.

Although our regression model produces continuous predictions, our action space is discrete (integer-valued actions). Therefore, we transform the continuous model outputs into discrete actions by simply selecting the nearest integer action to the model's predicted value. This rounding step introduces only a minor approximation error, and empirical analyses indicate that the effect is negligible in our context, given the granularity of the action grid.

1.2.1 Policy learning for linear shrinkage

There are (at least) two ways in which policy learning (PL) can improve the linear shrinkage estimator of the covariance matrix. The first approach, and arguably the more natural way to think about the problem, is to directly learn the optimal linear shrinkage intensity, that is:

$$\widehat{\Sigma}_{\delta^{\text{PL-L}}} := \widehat{\delta}^{\text{PL-L}} F + (1 - \widehat{\delta}^{\text{PL-L}}) S \ . \tag{1.14}$$

By construction, if $\widehat{\delta}^{\text{PL-L}}$ is larger (smaller) than $\widehat{\delta}$, our policy learning approach shrinks the sample covariance matrix more (less) toward the shrinkage target compared to the optimal shrinkage intensities from the literature derived under the i.i.d. assumption. Moreover, as long as $\widehat{\delta}^{\text{PL-L}} \in (0, 1]$, F is positive definite, and S is positive semidefinite (which holds for the sample covariance by construction), the estimator $\widehat{\Sigma}_{\delta^{\text{PL-L}}}$ is positive definite and therefore invertible.

One drawback of this approach is that it is not general enough to be applicable also to nonlinear shrinkage. We therefore reformulate the problem, no longer as a convex combination of the sample covariance matrix and a fixed target, but instead as a weighted combination of the sample eigenvalues and the shrunk eigenvalues (linear or nonlinear) proposed in the literature, as described in the next subsection.

1.2.2 Spectral policy learning for linear and nonlinear shrinkage

A more general approach, applicable to both linear and nonlinear shrinkage, is based on the spectral decomposition

$$\widehat{\Sigma}_{\gamma^{\text{PL-}*}} := U \Lambda_{\text{PL-}*} U' \ , \tag{1.15}$$

where $\Lambda_{\text{PL-}*} := \mathsf{Diag}(\lambda_1^{\text{PL-}*} \ldots, \lambda_N^{\text{PL-}*})$, and $* \in \{\text{L}, \ \text{NL}\}$ denotes if our estimator is applied to linear (I or CVC) or nonlinear shrunk eigenvalues. To estimate the optimal eigenvalues $\Lambda_{\text{PL-}*}$ of Equation (1.15), we propose the following convex combination of the sample eigenvalues and the * shrunk eigenvalues

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$$\tilde{\lambda}_i^{\text{PL-}*} := \widehat{\gamma}\lambda_i^* + (1-\widehat{\gamma})\lambda_i \ , \tag{1.16}$$

where $\widehat{\gamma} \in [0, \infty)$ is learned by the policy learning agent and $\tilde{\lambda}^{\text{PL-}*} := [\tilde{\lambda}_1^{\text{PL-}*}, \ldots, \tilde{\lambda}_N^{\text{PL-}*}]$. From Equation (1.16) it follows immediately that:

- the base shrinkage adjustment-factor $\widehat{\gamma}$ is constant[5];
- we reduce the estimation problem to $\mathcal{O}(1)$, while keeping the flexibility of nonlinear transformation, namely, learning the optimal convex combination of the sample eigenvalues and the *shrunk* eigenvalues proposed by the literature;
- for a "well-working" PL estimator, (i) $\widehat{\gamma} = 1$ indicates that the shrinkage estimator from the literature based on the i.i.d. assumption is well-suited for the investigated data, (ii) $\widehat{\gamma} > 1$ indicates that the eigenvalues need to be shrunk even more, (iii) $\widehat{\gamma} < 1$ indicates that the eigenvalues need to be shrunk less[6];
- $\tilde{\lambda}^{\text{PL-}*}$ is not guaranteed to be monotonic decreasing in the cases of $\widehat{\gamma} > 1$;
- and for too large $\widehat{\gamma}$, $\tilde{\lambda}^{\text{PL-}*}$ can have negative entries.

To guarantee that our policy learning covariance matrix estimator is well defined, that is, to maintain the (monotonic decreasing) order of the (shrunk) eigenvalues as well as their sign (only strictly positive eigenvalues such that the covariance matrix estimator is positive definite and invertible), we apply an isotonic-style correction to the spectrum. In general, isotonic regression refers to fitting a monotonic (typically non-decreasing) sequence to data. A standard formulation is to minimize the squared differences between fitted values and the data:

$$\lambda_i^{iso} = \ \text{argmin}_{\lambda \in \mathbb{R}^N} \sum_{i=1}^{N} (\lambda_i^{\text{PL-}*} - \lambda_i)^2 \quad \text{s.t.} \, \lambda_1 \geq \ldots \geq \lambda_N \ . \tag{1.17}$$

This projection reduces noise in the estimated eigenvalues while ensuring the structure required for a valid and well-conditioned covariance matrix. For a detailed overview on isotonic regression; see Barlow and Brunk (1972). Instead of solving this projection exactly, we apply a sequential transformation μ, iterating over $i = 1, \ldots, N$ (from largest to smallest eigenvalue), which provides an efficient approximation to the isotonic projection while preserving interpretability:

$$\lambda_i^{\text{PL-}*} = \mu\left(\tilde{\lambda}_i^{\text{PL-}*}, \lambda_i, \tilde{\lambda}^*\right) := \begin{cases} \lambda_1^{\text{PL-}*} := \max\left[\tilde{\lambda}_1^{\text{PL-}*}, \tilde{\lambda}^*\right] \\ \max\left[\min\left[\tilde{\lambda}_i^{\text{PL-}*}, \lambda_{i-1}^{\text{PL-}*}\right], \tilde{\lambda}^*\right] \quad \text{for} \ \dfrac{\lambda_i^*}{\lambda_i} < 1 \\ \min\left[\max\left[\tilde{\lambda}_i^{\text{PL-}*}, \lambda_{i-1}^{\text{PL-}*}\right], \tilde{\lambda}^*\right] \quad \text{for} \ \dfrac{\lambda_i^*}{\lambda_i} > 1 \end{cases} \tag{1.18}$$

where $\tilde{\lambda}^*$ denotes the crossing point to the 45-degree line (no shrinkage) in Figure 1, where we move from increasing the sample eigenvalues toward actually shrinking the sample

[5] Our PL approach can be applied also to individual shrinkage adjustments $\widehat{\gamma}_1, \ldots, \widehat{\gamma}_N$, but the number of degrees of freedom will increase to N and the probability to end up with an ill-conditioned estimator increases; for example, see the last two comments on monotonicity and invertability.

[6] By a 'well-working' PL estimator we mean an improved covariance matrix estimator $\widehat{\Sigma}_{\gamma\text{PL-}*}$ compared to $\widehat{\Sigma}_*$, measured by a lower out-of-sample standard deviation of the GMV portfolio as we explain later in Section 2.4.

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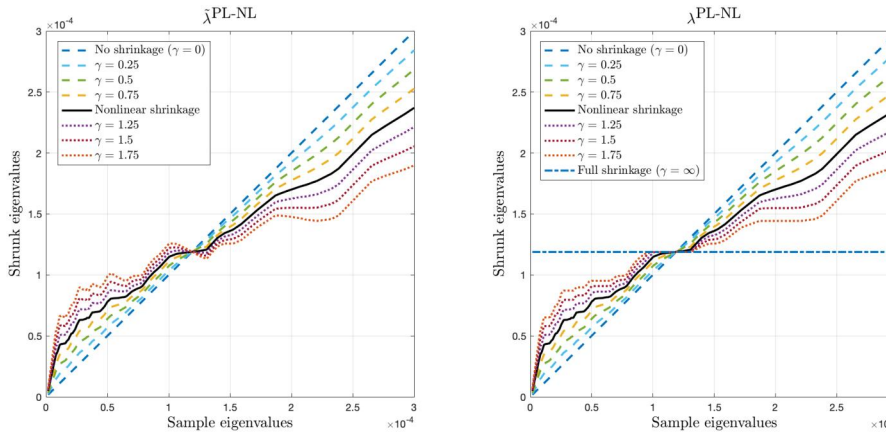


Figure 1. Illustration of the nonlinear shrinkage eigenvalue transformation shown in Equations (1.16) and (1.18) for various γ values. The panel on the left shows the raw transformation, whereas the panel on the right shows the monotonic decreasing shrunk eigenvalues after applying the function μ. The illustration is based on a representative sample as of January 08, 2014 for the $N = 100$ portfolio with $T = 252$.

eigenvalues. Thus $\tilde{\lambda}^* := (\lambda_j^* + \lambda_k^*)/2$ where j is the smallest (shrunk) eigenvalue that fulfills $\frac{\lambda_j^*}{\lambda_j} < 1$ and k is the largest (shrunk) eigenvalue that fulfills $\frac{\lambda_k^*}{\lambda_k} > 1$.

To visualize the impact of the proposed methodology on actual data, the first two figures plot the eigenvalues of a portfolio with 100 stocks based on daily stock return data, using a lookback window of one trading year. Further details are provided in Section 2.

In Figure 1, we visualize the impact of γ on the nonlinear shrunk eigenvalues (in black). A $\gamma \in [0, 1)$ rotates the original nonlinear shrinkage line counter-clockwise at the rotation point $(\bar{\lambda},\bar{\lambda})$, resulting in less shrinkage. A $\gamma > 1$ rotates the original nonlinear shrinkage line clockwise at the rotation point $(\bar{\lambda},\bar{\lambda})$, resulting in more shrinkage. The left panel of Figure 1 shows the raw transformations where the eigenvalues are not strictly monotonically decreasing, and the right panel shows the (Equation 1.18) adjusted transformation. A $\gamma = 0$ corresponds to the 45-degree line and therefore no shrinkage is applied. A $\gamma = \infty$ corresponds to the horizontal line and therefore maximal shrinkage to the grand sample eigenvalue mean, resulting in the identity matrix multiplied by the mean sample variance as covariance matrix estimator.

In Figure 2, we plot various (PL) transformed nonlinear shrinkage eigenvalues divided by their sample eigenvalues. First, more sample eigenvalues are pulled up than pushed down. Second, the smallest sample eigenvalues experience the largest impact, being disproportionately pulled up, whereas the largest sample eigenvalues are only slightly shrunk. Third, sample eigenvalues near their grand mean tend to be shrunk the least. Fourth, the largest and smallest sample eigenvalues are shrunk less than their "neighbors" closer to the mean. This implies that our PL estimator preserves some cross-sectional structure in the tail eigenvalues instead of flattening everything.

In Appendix A, we prove that for linear shrinkage toward a scalar multiple of the identity matrix, problem (Equation 1.14) is equivalent to problem (Equations 1.15–1.18). The intuition is that shrinking the sample covariance matrix toward the identity matrix (scaled by the average sample variance) is equivalent to shrinking the sample eigenvalues toward their grand mean. Consequently, from a mathematical perspective, learning the optimal convex combination $\widehat{\delta}^{\text{PL-I}}$—which corresponds to the slope of the straight line passing through the point $(\bar{\lambda},\bar{\lambda})$, with the sample eigenvalues on the x-axis and the shrunk

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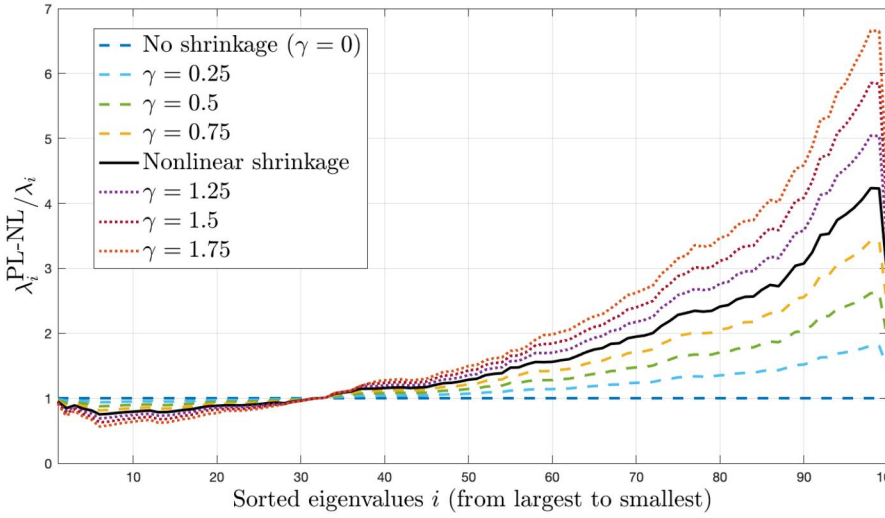


Figure 2. Illustration of the sorted eigenvalue transformations (Equation 1.18) for various γ values. The illustration is based on a representative sample as of January 08, 2014 for the $N = 100$ portfolio with $T = 252$.

eigenvalues on the y-axis, as shown in Figure 1—is equivalent to learning $\widehat{\gamma} \cdot \widehat{\delta}^{\text{I}}$, where $\widehat{\delta}^{\text{I}}$ is the classical slope from linear shrinkage and $\widehat{\gamma}$ is the slope adjustment determined by the policy learning agent.

Note that for other (linear) shrinkage targets, such as CVC, the sample eigenvalues are not shrunk toward their grand mean with the same intensity. As a result, problem (Equation 1.14) is not equivalent to problem (Equations 1.15–1.18). In unreported results, we find that both approaches outperform classical shrinkage based on the i.i.d. assumption. Moreover, we believe that learning the shrinkage intensity directly is a more natural approach. Finally, the monotonicity transformation in Equation (1.18) ensures that all shrunk eigenvalues remain strictly positive. Consequently, the covariance matrix estimator is positive definite and invertible.

1.3 Policy Class and Features

The action in our policy learning problem corresponds to choosing a shrinkage intensity. For linear shrinkage, we discretize the action space in $a \in \mathcal{A}_{\text{L}} := \{0, 1, \ldots, 100\}$, where $\widehat{\delta}^{\text{PL-L}} := a/100$, and thus $\widehat{\delta}^{\text{PL-L}} \in \{0, 0.01, 0.02, \ldots, 1\}$. For nonlinear shrinkage, we discretize the action space in $a \in \mathcal{A}_{\text{NL}} := \{0, 1, \ldots, 30\}$, where $\widehat{\gamma}^{\text{PL-NL}} := 0.5 + a/20$, and thus $\widehat{\gamma}^{\text{PL-NL}} \in \{0.5, 0.55, 0.60, \ldots, 2\}$.

In practice, the state space in a policy or RL framework can be flexibly specified. However, particularly in financial applications, it is crucial to carefully select state features that balance predictive power and noise. We construct our state space from a set of meaningful universe-specific signals measured at each trading date, summarized in Table 1. The "Linear Shrinkage" feature is solely used in our linear shrinkage estimator; else, the features are equal between the two models.

To approximate the optimal policy, we choose the elastic net (ENet) as a regularized regression method over more complex deep learning methods due to the relatively small number of actions and the moderate dimensionality of the state space. ENet provides robustness to noise, interpretability, and inherent regularization, which effectively mitigates overfitting; a common problem of financial backtests. Additionally, its simplicity helps to

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Table 1. Summary of market state features, computed over 1-year estimation windows based on the eligible stock universe.

Feature	Description
Average correlation	Average pairwise sample correlation between stocks.
Average volatility	Average one-year sample volatility of all individual stocks.
EWMA	Exponentially weighted moving average of previous month returns (decay-$=0.1$) of the equally weighted portfolio.
Lagged optimal action	Optimal action from previous rebalancing.
Lagged optimal SD	Standard deviation corresponding to lagged optimal action.
Linear shrinkage	Linear shrinkage intensity of Ledoit and Wolf (2004b).
Momentum	Fraction of days each stock had positive returns over the previous month, averaged across all stocks.
Rolling optimal action	One-year rolling average of optimal action.
Rolling optimal SD	One-year rolling standard deviation corresponding to optimal actions.
Trace	Trace of sample covariance matrix.
Universe volatility	One-year sample volatility of the equally weighted universe.

avoid the computational complexity and instability often associated with deep learning methods when applied to limited and noisy financial datasets.

Our model is trained on approximately 40 years of historical financial data, with roughly half of this dataset (approximately 20 years) used for initial training. For each trading day, we evaluate all available actions (101 for PL-L and 31 for PL-NL) by calculating the 21-days ahead portfolio standard deviation. The optimal action, defined as the one that minimizes this realized risk measure, becomes our target variable.

We conduct in-sample hyperparameter tuning over the grid summarized in Table 2, noting that predictive performance remains robust and largely unchanged across this grid. The optimized ENet hyperparameters are defined as follows:

- α: Controls overall regularization strength. Higher α implies stronger regularization.
- **l1_ratio**: Controls the balance between L_1 and L_2 penalties. Specifically, $l1_ratio = 0$ results in pure L_2 regularization, $l1_ratio = 1$ results in pure L_1 regularization, and intermediate values provide a balance.
- **max_iter**: Maximum iterations allowed for optimization convergence.
- **tol**: Convergence tolerance; optimization stops once parameter updates become smaller than this value.

Additionally, we employ a rolling-window retraining scheme, updating our model every 21 trading days, corresponding to each rebalancing period, by incorporating the most recent 21 observations and discarding the oldest 21 observations. This rolling-window approach ensures that our model adapts continually to the latest market conditions, providing predictions closely aligned with current market regimes. This is another benefit of employing a simpler, less computationally intensive model.

Remark 1.1 (Limitations of the Policy Learning Approach).
In low signal-to-noise environments with an i.i.d. data-generating process, the incremental value of the proposed policy learning approach is limited to none relative to classical shrinkage estimators, and the added complexity is therefore hard to justify. The approach can also be vulnerable to abrupt regime switches and to unobserved or rare events that are absent from the training sample and thus cannot be learned. Nevertheless, because the learned shrinkage intensity is bounded ($\widehat{\delta}^{\text{PL-L}} \in [0,1]$ and $\widehat{\gamma}^{\text{PL-NL}} \in [0.5,2]$) and typically avoids extreme values (see

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Table 2. Hyperparameter grid for ENet regression.

Hyperparameter	Values
α	0.5, 1.0, 1.5, 2.0, 5.0
l1_ratio	0.1, 0.25, 0.5, 0.75, 0.9
max_iter	500, 1000, 1500, 2000
tol	$10^{-3}, 10^{-4}, 10^{-5}$

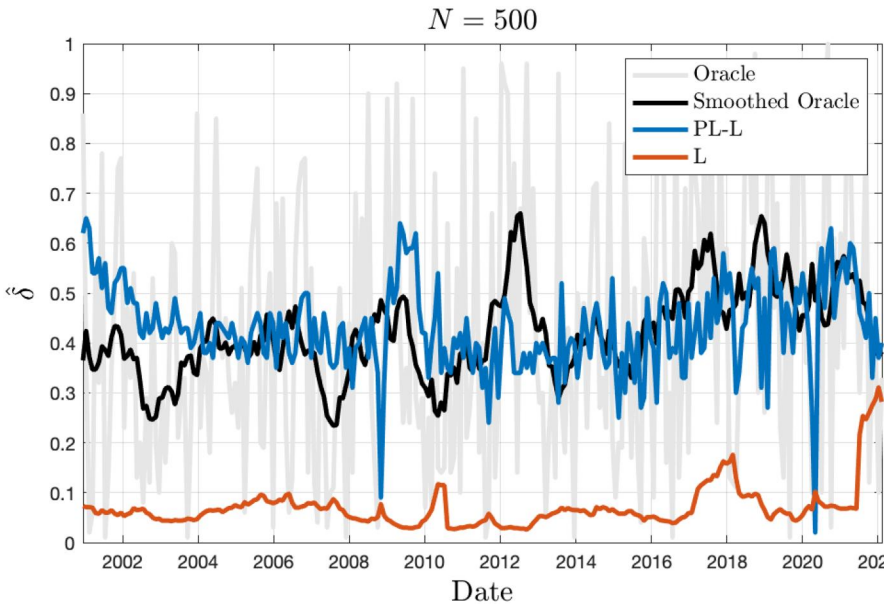


Figure 3. Comparison of the monthly Oracle, PL-L, and L estimated linear shrinkage intensities over time for $N = 500$. For the smoothed Oracle, we use a 1-year moving average smoothing.

Figures 3 and 4), it yields a well-defined estimator and continues to deliver reasonable results under these stressors—though not necessarily improvements over traditional shrinkage estimators.

Remark 1.2 (Adjustments for Long-Only Portfolios).

As we point out later, long-only portfolios need less shrinkage. This is why in the PL-NL long-only case we extend the action space by $a \in \mathcal{A}_{\text{NL-long-only}} := \{0, 1, \ldots, 40\}$, where $\widehat{\gamma}^{\text{PL-NL-long-only}} := a/20$, and thus $\widehat{\gamma}^{\text{PL-NL-long-only}} \in \{0, 0.05, 0.10, \ldots, 2\}$.
In our long-only portfolio context, we additionally employ a cost-sensitive learning approach, biasing the model toward predicting lower actions and thereby less shrinkage. In our case, this is an inverse logarithmic weighting $\frac{1}{log(1+a)}$ of the actions in the training set, weighting lower actions, and therefore smaller shrinkage intensities, more. This is based on our in-sample findings indicating that lower actions generally yield lower standard deviations. This empirical evidence aligns with prior research, which indicates that long-only portfolios generally require

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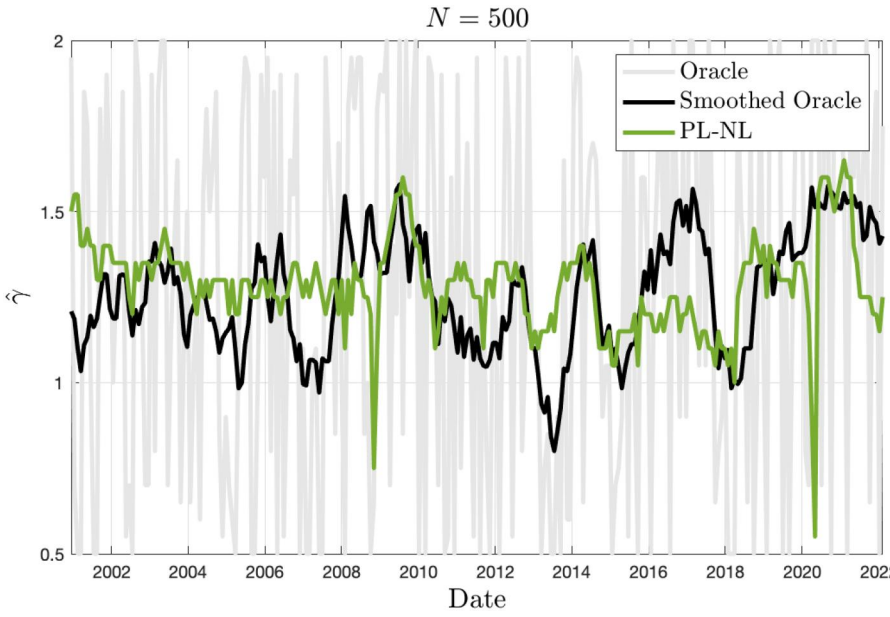


Figure 4. Comparison of the Oracle and PL-NL estimated nonlinear shrinkage intensities adjustment over time for $N = 500$. For the smoothed Oracle, we use a 1-year moving average smoothing.

substantially less regularization compared to their unconstrained (long-short) counterparts (see, e.g., Jagannathan and Ma 2003; Dom et al. 2025).

2 Empirical Analysis

2.1 Data and General Portfolio-Construction Rules

We download daily stock return data from the Center for Research in Security Prices (CRSP) starting on January 1, 1980 through January 31, 2022. We restrict attention to stocks from the NYSE, AMEX, and NASDAQ stock exchanges.

Roughly half of the data is allocated for training our PL shrinkage estimators, with the remaining half used for evaluation. Hence, our out-of-sample period spans from December 18, 2000 to January 31, 2022. This results in a total of 253 months or 5313 trading days for evaluation. All portfolios are updated monthly.[7] For simplicity, and in line with much of the literature, we adopt the common convention that 21 consecutive trading days constitute one (trading) "month." This implies that the covariance matrix must be estimated every 21 trading days, that is, every rebalancing date $h = 1, \ldots, 253$. At any investment date h, a covariance matrix is estimated based on the most recent 252 daily returns, which roughly corresponds to using 1-year of past data.

We consider the following portfolio sizes: $N \in \{30, 50, 100, 225, 500\}$. For a given combination (h, N), the investment universe is obtained as follows. We find the set of stocks that have an almost complete return history over the most recent $T = 252$ days as well as a complete return "future" over the next 21 days.[8]

[7] Monthly updating is common practice to avoid an unreasonable amount of turnover and thus transaction costs. During a month, from one day to the next, we hold number of shares fixed rather than portfolio weights; in this way, there are no transactions during a month.

[8] The first restriction allows for up to 2.5% of missing returns over the most recent 252 days, and replaces missing values by zero. The latter, 'forward-looking' restriction is not feasible in practice but is commonly used

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From the remaining set of stocks, we then pick the largest N stocks (as measured by their market capitalization on investment date h) as our investment universe. In this way, the investment universe changes relatively slowly from one investment date to the next.

There is a great advantage in having a well-defined rule that does not involve drawing stocks at random, as such a scheme would have to be replicated many times and averaged over to give stable results. As far as rules go, the one we have chosen seems the most reasonable because it avoids so-called "penny stocks" whose behavior is often erratic; also, high-market-cap stocks tend to have the lowest bid-ask spreads and the highest depth in the order book, which allows large investment funds to invest in them without breaching standard safety guidelines.

2.2 Competing Covariance Matrix Estimators

We now detail the various covariance matrix estimators included in our empirical analysis.

- **L:** the **l**inear shrinkage estimator of Ledoit and Wolf (2004a); that is, we use the scalar multiple of the identity matrix (I) shrinkage target.
- **NL:** the **n**onlinear shrinkage estimator of Ledoit and Wolf (2022b), that is, we use the quadratic-inverse shrinkage (QIS) estimator.
- **PL-L:** the **l**inear shrinkage estimator of Ledoit and Wolf (2004a), but with data-driven shrinkage intensity computed via **p**olicy **l**earning as described in Section 1.2.1.
- **PL-NL:** the **n**onlinear shrinkage estimator of Ledoit and Wolf (2022b), but with data-driven shrinkage intensities computed via **p**olicy **l**earning as described in Section 1.2.2.

It is of interest to include the sample covariance matrix and other linear and nonlinear shrinkage estimators from the literature to compare their performance. However, as summarized in Ledoit and Wolf (2022a), nonlinear shrinkage estimators tend to perform similarly, whereas the sample covariance matrix is consistently outperformed by both linear and nonlinear shrinkage estimators. We confirm these findings and, to save space, do not report (all) results for the sample covariance matrix, instead focusing on the QIS nonlinear shrinkage estimator. For linear shrinkage, we present results only for the identity-based shrinkage target (I), as it is the only case where both policy learning approaches yield identical results: (i) learning the optimal convex combination δ of the sample covariance matrix and the shrinkage target, as detailed in Equation (1.14), or (ii) learning the optimal convex combination γ of the sample eigenvalues and the shrunk eigenvalues, as described in Equation (1.16); see Appendix A. Nonetheless, in unreported results, we find that for more sophisticated linear shrinkage targets—such as the constant-variance-covariance shrinkage target of De Nard (2022) in Equation (1.8)—the results discussed in Section 2.5 hold as well and generally show improved performance for both approaches.

2.3 Performance Measures

- **AV:** We compute the **av**erage of the 5313 out-of-sample returns as geometric mean, and then multiply by $\sqrt{252}$ to annualize.
- **SD:** We compute the **s**tandard **d**eviation of the 5313 out-of-sample returns, and then multiply by $\sqrt{252}$ to annualize.
- **SR:** We compute the annualized **S**harpe **r**atio as the ratio AV/SD.
- **TO:** We compute average (monthly) **t**urn**o**ver as $\frac{1}{252}\sum_{h=1}^{252}||\widehat{w}_{h+1}-\widehat{w}_h^{\text{hold}}||_1$, where $||\cdot||_1$ denotes the L_1 norm and $\widehat{w}_h^{\text{hold}}$ denotes the vector of the "hold" portfolio weights at the end of month h.[9]

in the literature. Although it might affect (in a minor way) absolute performance due to survivorship bias, it does not systematically affect relative performance of various methods.

[9] The vector $\widehat{w}_h^{\text{hold}}$ is determined by the initial vector of portfolio weights, $\widehat{w}_h$, together with the evolution of the various prices of the N stocks in the portfolio during month h.

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- **GL:** We compute average (monthly) excess **g**ross **l**everage as $\frac{1}{253}\sum_{k=1}^{253}||\widehat{w}_h||_1 - 1$.
- **PL:** We compute average (monthly) **p**roportion of **l**everage as $\frac{1}{253\times N}\sum_{h=1}^{253}\sum_{i=1}^{N}\mathbb{1}_{\{\widehat{w}_{i,h}<0\}}$.

2.4 GMV Portfolio

We consider the problem of estimating the GMV portfolio in the absence of short-sales constraints. The problem is formulated as

$$\arg\min_{w} w'\Sigma w \tag{2.1}$$

$$\text{subject to } w'\mathbb{1} = 1\ , \tag{2.2}$$

where $\mathbb{1}$ denotes a vector of ones of dimension $N\times 1$. It has the analytical solution

$$w = \frac{\Sigma^{-1}\mathbb{1}}{\mathbb{1}'\Sigma^{-1}\mathbb{1}}\ . \tag{2.3}$$

The natural strategy in practice is to replace the unknown covariance matrix Σ by an estimator $\widehat{\Sigma}$ in Equation (2.3), yielding a feasible portfolio

$$\widehat{w} := \frac{\widehat{\Sigma}^{-1}\mathbb{1}}{\mathbb{1}'\widehat{\Sigma}^{-1}\mathbb{1}}\ . \tag{2.4}$$

Estimating the GMV portfolio is a "clean" problem in terms of evaluating the quality of a covariance matrix estimator, as it abstracts from having to estimate the vector of expected returns at the same time. In addition, researchers have established that estimated GMV portfolios have desirable out-of-sample properties not only in terms of risk but also in terms of reward-to-risk, that is, in terms of Sharpe and information ratios (see, e.g., Haugen and Baker 1991; Jagannathan and Ma 2003; Nielsen and Aylursubramanian 2008).

Following the advice of Dom et al. (2025) and focusing on more practically relevant portfolios, we also include an analysis of long-only minimum variance portfolios for investors who cannot take short positions. Notably, although the covariance matrix estimator in classical shrinkage is independent of the specific investment problem, for PL shrinkage, it plays a crucial role. The PL agent learns the optimal shrinkage intensity based on the investment environment and portfolio constraints, which means that the intensity for long-short and long-only portfolios can differ. As we will demonstrate, they indeed do.

In the long-only case, the weights of the GMV portfolio need to be derived numerically. This is a constrained quadratic optimization problem, which we solve with the python package *CVXPY*.

In addition to Markowitz portfolios based on Equation (2.4), we also include as a simple-minded benchmark the equal-weighted portfolio promoted by DeMiguel, Garlappi, and Uppal (2009), among others, as it has been claimed to be difficult to outperform. We denote the equal-weighted portfolio by **EW**. We also include the value-weighted portfolio, denoted by **VW**, which is sometimes favored over the equal-weighted portfolio, for example because it incurs less turnover and "controls" for small-cap biases.

Our stance is that in the context of the GMV portfolio, the most important performance measure is the out-of-sample standard deviation, SD. The true (but unfeasible) GMV portfolio is given by Equation (2.3). It is designed to minimize the variance (and thus the standard deviation) rather than to maximize the expected return or the information ratio. Therefore, any portfolio that implements the GMV portfolio should be primarily evaluated

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by how successfully it achieves this goal. A high out-of-sample average return, AV, and a high out-of-sample Sharpe ratio, SR, are naturally also desirable, but should be considered of secondary importance from the point of view of evaluating the quality of a covariance matrix estimator.

We also consider the question of whether one estimation model delivers a lower out-of-sample standard deviation than another estimation model. To avoid a multiple testing problem and as a major goal of this study is to show that using a data-driven RL approach to obtain the shrinkage intensities improves the estimation of (large-dimensional) covariances matrices, we restrict attention to two comparisons: (i) L with PL-L, and (ii) NL with PL-NL. For a given universe size, a two-sided p-value for the null hypothesis of equal standard deviations is obtained by the prewhitened HAC_{PW} method described in Ledoit and Wolf (2011, Section 3.1).[10]

2.5 Empirical Results

2.5.1 Main results

The main results are presented in Table 3 and can be summarized as follows. Unless stated otherwise, the findings are with respect to SD as the performance measure.

- All models outperform EW and VW by a wide margin.
- Each PL-L and PL-NL model consistently—and often markedly—outperforms its respective classical L and NL base model. Moreover, the outperformance of PL-L over L, and PL-NL over NL, is both statistically significant and economically meaningful.
- There is a consistent ranking across smaller portfolio sizes $N \in \{30, 50, 100\}$ (from best to worst): PL-L, PL-NL, NL, L. For larger portfolio sizes $N \in \{225, 500\}$, the ranking shifts to (from best to worst): PL-NL, NL, PL-L, L, VW, EW.

To sum up, the proposed policy learning approach for computing shrinkage intensity (ies) outperforms classical linear and nonlinear shrinkage estimators based on the i.i.d. assumption. Consistent with prior theory and evidence, linear shrinkage is typically preferred for smaller cross-sections (roughly $N \leq 100$), whereas the gains from nonlinear shrinkage materialize as N becomes large (e.g., see Ledoit and Wolf 2017; De Nard 2022). Our empirical results closely mirror these patterns.

To make effect sizes more interpretable, we report the proposed estimators relative to EW and VW. Specifically, Table 4 reports the standard deviations from Table 3 divided by the corresponding EW and VW SD's. Across portfolio sizes N, both the classical shrinkage and PL-based portfolios deliver *ratios well below 1.0*, making the improvements over naïve benchmarks transparent in absolute and relative terms. Moreover, PL-L markedly improves upon L—by between 50 and 600 percentage points (pp) for EW, and between 50 and 610 pp for VW—while the added value of PL-NL over NL is consistent and robust, though smaller (10–40 pp for both EW and VW). For completeness, we also report PL-L/L and PL-NL/NL ratios to quantify the incremental value of the policy learning enhancement over the corresponding Ledoit–Wolf baselines and find similar results: PL-L reduces SD's by 1%–10% relative to L, whereas the reduction for PL-NL is below 1%.

We also report a comparison over time of the classical linear shrinkage intensity estimates and those learned by PL in Figure 3 for $N = 500$. For other portfolio sizes, see Figure B.1. For L, the shrinkage intensity is very low and stable, with an average shrinkage intensity of 0.07. This is due to the fact that the scalar multiple of the identity matrix as a

[10] As the out-of-sample size is very large at 5,313, there is no need to use the computationally more involved bootstrap method described in Ledoit and Wolf (2011, Section 3.2), which is preferred for small sample sizes.

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Table 3. Annualized out-of-sample standard deviations for the competing estimators of the GMV portfolio, in percentage.

SD of the global minimum variance portfolio						
	L	PL-L	NL	PL-NL	EW	VW
$N = 30$	13.50	**13.41**	13.45	13.42	18.72	19.13
$N = 50$	13.64	13.28*	13.32	**13.25***	18.86	19.04
$N = 100$	13.26	**12.57***	12.68	12.62*	19.51	19.20
$N = 225$	12.64	11.45*	11.42	**11.35***	19.87	19.32
$N = 500$	11.21	10.45*	10.38	**10.36**	20.62	19.52

All measures are based on 5313 daily out-of-sample returns from December 18, 2000 until January 31, 2022. For any portfolio size, the best estimator is indicated in **bold** font. In the columns PL-L, respectively PL-NL, significance at the 0.01 level of outperformance in SD over L, respectively NL, is denoted by an asterisk (*).

Table 4. All entries are ratios of SD's from Table 3 relative to (i) their L, respectively NL, benchmarks (BM); (ii) relative to EW SD's; and (iii) relative to VW SD's.

SD ratios of global minimum variance portfolios										
	PL-L	PL-NL	L	PL-L	NL	PL-NL	L	PL-L	NL	PL-NL
	Divided by BM		Divided by EW				Divided by VW			
$N = 30$	0.993	0.998	0.721	0.716	0.718	0.717	0.706	0.701	0.703	0.702
$N = 50$	0.974	0.995	0.723	0.704	0.706	0.703	0.716	0.697	0.700	0.696
$N = 100$	0.948	0.995	0.680	0.644	0.650	0.647	0.691	0.655	0.660	0.657
$N = 225$	0.906	0.994	0.636	0.576	0.575	0.571	0.654	0.593	0.591	0.587
$N = 500$	0.932	0.998	0.544	0.507	0.503	0.502	0.574	0.535	0.532	0.531

shrinkage target has very high bias but very low estimation error.[11] In contrast, the purely data-driven PL-L approach yields much higher and more adaptive shrinkage intensities, with an average shrinkage intensity of 0.43. Comparing these shrinkage intensities with the (1-year moving average of the) Oracle—defined as the best ex-post monthly shrinkage intensity that minimizes monthly out-of-sample SD—we observe that PL-L successfully learns a higher level of shrinkage, closely aligning with the smoothed Oracle (average shrinkage intensity of 0.42). Overall, we find that classical linear shrinkage based on the i.i.d. assumption results in a substantially lower shrinkage intensity, which can be consistently improved by PL-L.

A similar but less extreme pattern can also be observed for nonlinear shrinkage. In Figure 4, we plot the adjustment of nonlinear shrinkage intensities over time for $N = 500$; for other portfolio sizes, see Figure B.2. Since $\widehat{\gamma} = 1$ corresponds to classical nonlinear shrinkage, we find that PL-NL consistently exhibits substantially higher shrinkage intensities. This suggests that sample eigenvalues require even greater adjustment.

This observation is further supported by the Oracle nonlinear shrinkage intensities, which are generally well above 1. In particular, their smoothed version (with an average adjustment of 1.26) closely aligns with the PL-NL values (average adjustment of 1.28).

Notably, PL shrinkage intensities spike during periods of financial turmoil—such as the dot-com bubble, the global financial crisis, and the COVID-19 pandemic—indicating that the need for shrinkage is highest when market volatility is elevated.

[11] See De Nard (2022). Note that for other linear shrinkage estimators, bias can be reduced by incorporating a more realistic shrinkage target, which increases the optimal shrinkage intensity. However, more realistic shrinkage targets require estimating additional (and more complex) parameters, leading to higher estimation errors. As a result, the estimated optimal shrinkage intensities become more volatile.

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To visualize the monthly out-of-sample shrinkage intensity estimation errors, we present the mean, mean absolute, and mean squared deviations of the oracle intensities minus the estimated intensities in Table 5 and Figure 5. The estimation error results can be summarized as follows:

Table 5. Monthly shrinkage-intensity estimation errors of the GMV portfolio.

Monthly shrinkage intensity estimation errors of GMV portfolios												
	Mean error				Mean absolute error				Mean squared error			
	L	PL-L	NL	PL-NL	L	PL-L	NL	PL-NL	L	PL-L	NL	PL-NL
$N = 30$	0.21	−0.01	0.35	0.04	0.25	0.24	0.65	0.58	0.13	0.08	0.52	0.40
$N = 50$	0.25	0.00	0.37	0.00	0.28	0.24	0.63	0.56	0.15	0.08	0.51	0.37
$N = 100$	0.29	0.02	0.41	0.06	0.30	0.23	0.67	0.57	0.16	0.08	0.54	0.39
$N = 225$	0.36	0.02	0.43	0.07	0.37	0.23	0.60	0.49	0.20	0.08	0.47	0.30
$N = 500$	0.35	−0.01	0.26	−0.02	0.36	0.23	0.53	0.49	0.20	0.08	0.37	0.30

All numbers are based on 5313 daily out-of-sample returns, respectively 253 monthly predicted shrinkage intensities versus the oracle intensities, from December 18, 2000 until January 31, 2022.

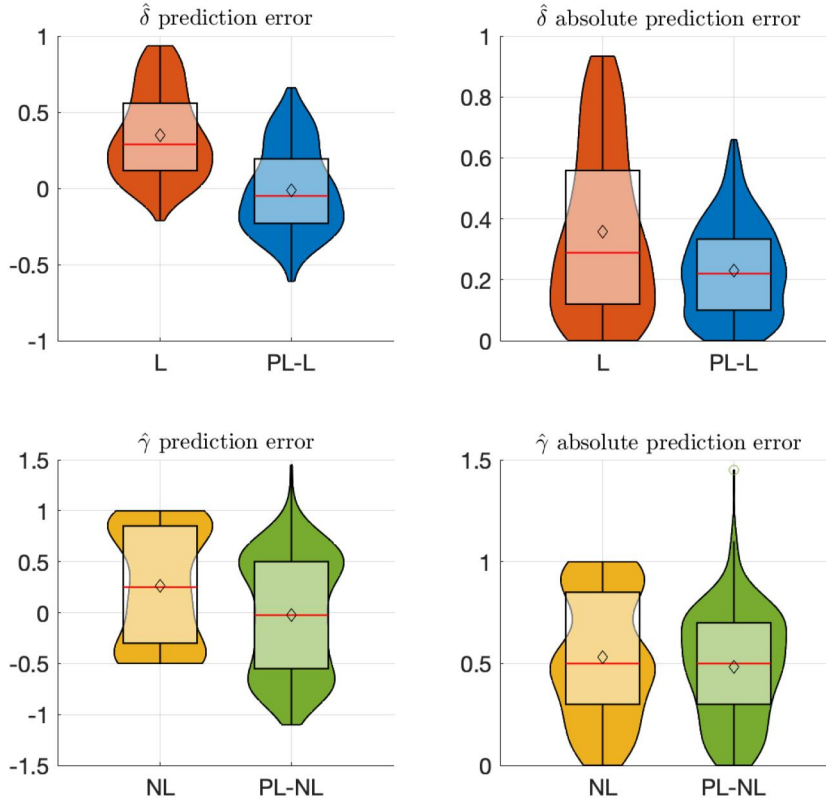


Figure 5. Violin plots of monthly Oracle versus estimated optimal shrinkage intensity differences (panels on the left) and absolute differences (panels on the right) for the $N = 500$ GMV portfolio. The upper panels show the differences for linear shrinkage and the lower panels for nonlinear shrinkage. A violin plot visually summarizes the distribution of a data set by combining the corresponding box plot with a (rotated) kernel density estimator; note that within the box of the box plot the sample median is indicated by a horizontal line whereas the sample average is indicated by a diamond.

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- For any portfolio size and deviation metric, PL shrinkage intensities exhibit consistently and significantly lower estimation errors compared to classical shrinkage intensities.
- The positive mean error values for L and NL indicate a substantial negative bias in shrinkage intensities due to the i.i.d. assumption.
- In contrast, the mean errors of the data-driven PL shrinkage intensities are close to zero.

In Figure B.3, we plot the correlation of linear shrinkage intensities across all models and find that they are positively correlated. Notably, we observe two distinct clusters of models with high correlations: (i) The classical L models across N, and (ii) the new PL-L models across N. The correlations between L and PL-L are substantially lower, indicating that the data-driven approach not only increases the level of shrinkage but also alters its dynamics. This is further evident in Figures 3 and B1.

In unreported results, we find that the added value of PL extends beyond merely reducing the negative bias of the estimated shrinkage intensities (i.e., increasing the shrinkage level). It also enhances the shrinkage dynamics. To test this, we adjust the linear shrinkage intensities by adding the bias term (first L column of Table 5).

To illustrate the impact of the shrinkage level, we also report the standard deviation of the GMV portfolio for constant shrinkage intensities in Table 6:

- All classical and PL shrinkage models consistently outperform the sample covariance matrix S (i.e., $\delta = \gamma = 0$). The SD of S increases with N due to greater estimation error in higher dimensions, whereas shrinkage models reduce SD by improving estimation accuracy and enhancing diversification in higher dimensions.
- Total shrinkage results in the identity matrix as the covariance matrix estimator, which is equivalent to the EW portfolio ($\delta = 1$ or $\gamma = \infty$).
- Consequently, both excessively small and excessively large shrinkage levels are suboptimal. We find that the optimal constant linear shrinkage intensity increases with N due to the curse of dimensionality.
- For linear shrinkage toward the identity matrix, the optimal constant shrinkage intensity consistently outperforms L and is comparable to PL-L. This suggests that L can be improved by increasing its shrinkage intensity. However, since the optimal constant shrinkage intensity is determined ex post, it is not directly applicable, whereas PL-L is.
- For nonlinear shrinkage, we observe that the shrunk eigenvalues require even stronger shrinkage, as optimal γ values are greater than one. However, the optimal constant γ adjustment decreases with N.
- The SDs of the ex post evaluated optimal constant γ adjustments are comparable to those of PL-NL.

Additionally, we report results on average turnover and leverage. The results, presented in Table 7, can be summarized as follows. Note that our optimization does not impose constraints on leverage or turnover.

- For linear shrinkage, PL consistently and significantly reduces turnover. Overall, PL-L exhibits the lowest turnover across all models.
- For linear shrinkage, PL also results in lower leverage for larger portfolio sizes.
- Turnover and leverage values are comparable between NL and PL-NL.

In Figure 6, we report the average variable-importance scores for PL-L and PL-NL over the entire out-of-sample period for $N = 500$. In line with de Prado (2018), we measure feature importance using mean decrease impurity (MDI), which attributes to each feature the

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Table 6. Annualized out-of-sample standard deviations of the GMV portfolio for the linear (l) and nonlinear (QIS) shrinkage estimators with fixed δ respectively γ.

Linear shrinkage											
δ	**0.0**	**0.1**	**0.2**	**0.3**	**0.4**	**0.5**	**0.6**	**0.7**	**0.8**	**0.9**	**1.0**
$N = 30$	13.64	**13.35**	13.36	13.49	13.70	13.99	14.37	14.88	15.59	16.68	18.72
$N = 50$	13.96	13.35	**13.24**	13.29	13.42	13.64	13.94	14.37	15.04	16.19	18.86
$N = 100$	14.27	12.80	**12.60**	12.62	12.73	12.91	13.19	13.59	14.23	15.49	19.51
$N = 225$	26.50	11.75	11.41	**11.39**	11.48	11.67	11.93	12.32	12.91	14.08	19.87
$N = 500$	NA	10.73	10.46	**10.43**	10.50	10.65	10.89	11.24	11.79	12.87	20.62
Nonlinear shrinkage											
γ	**0.5**	**0.6**	**0.7**	**0.8**	**0.9**	**1.0**	**1.2**	**1.4**	**1.6**	**1.8**	**2.0**
$N = 30$	13.53	13.50	13.49	13.47	13.46	13.45	13.42	13.42	13.38	**13.38**	13.39
$N = 50$	13.52	13.46	13.41	13.38	13.37	13.32	13.27	13.24	13.20	**13.19**	13.18
$N = 100$	13.00	12.90	12.83	12.76	12.72	12.68	12.64	**12.61**	12.62	12.63	12.65
$N = 225$	11.88	11.72	11.61	11.52	11.46	11.42	11.36	**11.34**	11.34	11.36	11.42
$N = 500$	10.62	10.53	10.47	10.43	10.40	10.38	**10.37**	10.39	10.46	10.54	10.62

All measures are based on 5313 daily out-of-sample returns from December 18, 2000 until January 31, 2022. For both estimators and for any portfolio size the best fixed δ, respectively γ, is indicated in **bold** font.

Table 7. Monthly average portfolio metrics for the competing estimators of the GMV portfolio.

Global minimum variance portfolio						
	l	**PL-l**	**NL**	**PL-NL**	**EW**	**VW**
			TO			
$N = 30$	0.61	**0.09**	0.80	0.80	0.06	0.07
$N = 50$	0.94	**0.16**	0.86	0.86	0.06	0.07
$N = 100$	1.60	**0.29**	1.02	1.11	0.06	0.07
$N = 225$	3.03	**0.54**	1.90	1.79	0.05	0.06
$N = 500$	3.10	**1.07**	3.55	3.59	0.05	0.06
			GL			
$N = 30$	2.05	3.82	2.20	**2.15**	1.00	1.00
$N = 50$	2.61	3.86	2.36	**2.30**	1.00	1.00
$N = 100$	3.62	3.62	2.67	**2.60**	1.00	1.00
$N = 225$	5.59	**3.25**	4.02	3.97	1.00	1.00
$N = 500$	5.67	**2.96**	4.99	5.09	1.00	1.00
			PL			
$N = 30$	0.35	0.41	0.33	**0.33**	0.00	0.00
$N = 50$	0.38	0.41	0.35	**0.34**	0.00	0.00
$N = 100$	0.43	0.41	0.38	**0.37**	0.00	0.00
$N = 225$	0.45	**0.39**	0.42	0.42	0.00	0.00
$N = 500$	0.44	**0.38**	0.43	0.43	0.00	0.00

All measures are based on 253 monthly out-of-sample weight vectors December 18, 2000 until January 31, 2022. The best estimator for any measure (TO, GL, and PL) is indicated in **bold** font. The EW and VW portfolio are not considered in this comparison.

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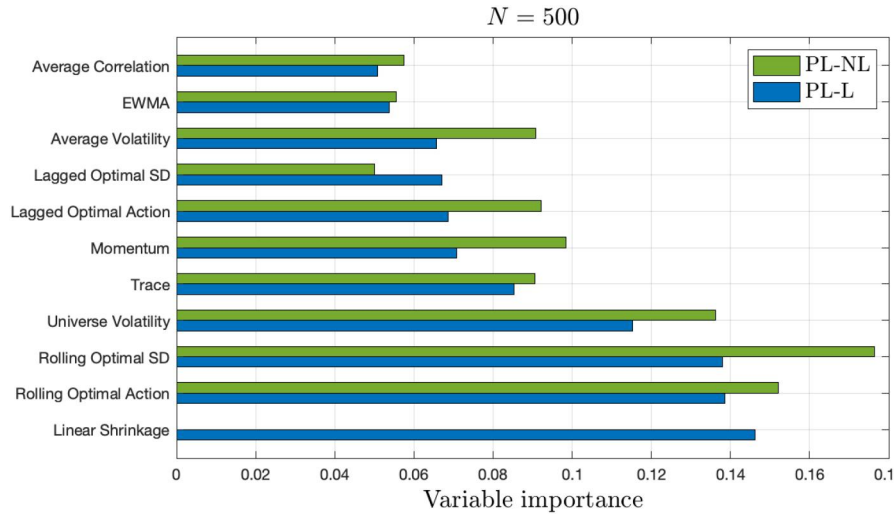


Figure 6. Average variable importance across 253 monthly fits of PL-L and PL-NL for $N = 500$.

average reduction in node impurity it achieves across all trees in a random forest. The main findings are:

- Unsurprisingly, for PL-L the linear shrinkage intensity of Ledoit and Wolf (2004b) emerges as the most important feature, effectively serving as a prior/anchor for the policy.
- Both the rolling optimal action (shrinkage intensity) and its rolling standard deviation rank among the most influential variables.
- Universe-level volatility also carries substantial importance, helping the policy learner to identify how much shrinkage is appropriate.

Overall, the variable-importance profiles for PL-L and PL-NL are similar and remain comparable across N.

2.5.2 Robustness checks

Table 3 presents the "single" results over the entire out-of-sample period from December 18, 2000 to January 31, 2022. A natural question arises: is the relative performance of the various models stable throughout this period, or does it change during certain subperiods, such as periods of "boom" or "bust" compared to normal conditions? To investigate this, we conduct a rolling-window analysis based on shorter out-of-sample periods.

The results for out-of-sample standard deviations using a one-year rolling window are displayed in Figure 7. The figure shows results for linear shrinkage GMV portfolios with a universe size of $N = 500$. The relative performance appears remarkably stable over time, with PL-L consistently performing best across all subperiods. Specifically, PL-L outperforms L in 95% of trading days based on one-year rolling SDs. Moreover, PL-L remains very close to the Oracle performance for most of the period, whereas L consistently performs markedly worse than the theoretical optimum.

Interestingly, the advantage of PL over classical shrinkage intensities remains largely time-invariant: we do not observe systematic differences in PL's outperformance during

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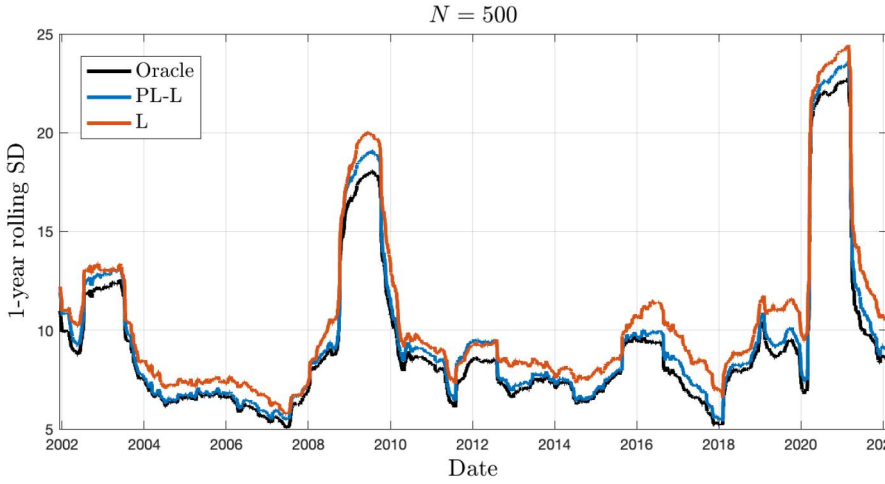


Figure 7. One-year rolling-window of out-of-sample standard deviations (SDs) for linear shrinkage models based on the $N = 500$ GMV portfolio.

periods of "boom" or "bust" compared to normal periods. This finding reinforces the robustness and stability of PL's improvements.

To save space and avoid redundancy, we do not report results for nonlinear shrinkage. However, the same pattern of robust outperformance over time is observed for PL-NL versus NL. That said, the relative improvement of PL is smaller for nonlinear shrinkage, as also shown in Tables 3 and 4.

For linear shrinkage, we also examine alternative shrinkage targets proposed in the literature, such as CVC. In unreported results, we find that the PL version consistently improves upon classical linear shrinkage across various shrinkage targets. Although the reduction in SD is statistically significant, the relative improvement tends to be smaller for more sophisticated shrinkage targets. This is because their shrinkage intensity is often substantially larger due to their enhanced accuracy; see De Nard (2022).[12]

Additionally, we conduct various robustness checks on the number of stocks in the portfolio (N), the number of observations in the estimation window (T), and, consequently, different concentration ratios. The results confirm that our findings are robust and not driven by specific parameter choices.

2.5.3 Long-only minimum variance portfolios

In this section, we present the results for long-only minimum variance portfolios. While classical shrinkage covariance matrix estimators are independent of the portfolio optimization step, the ex post optimal shrinkage intensity (Oracle) differs, which also affects the learned shrinkage intensity of the PL agent. The long-only results are summarized in Table 8 and visualized in Figures 8 and 9.

- Each PL-L and PL-NL model consistently outperforms its respective classical L and NL base model. For NL and higher-dimensional settings, PL-NL's outperformance is both statistically significant and economically meaningful.
- A consistent ranking emerges for smaller portfolio sizes $N \in \{30, 50, 100\}$, ordered from best to worst as: PL-L, L, PL-NL, NL. For larger portfolio sizes $N \in \{225, 500\}$, the ranking shifts to: PL-NL, NL, PL-L, L, VW, EW.

[12] The results are available upon request.

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Table 8. Annualized out-of-sample standard deviations for the competing estimators of the long-only minimum variance portfolio, in percentage.

SD of the long-only minimum variance portfolio						
	L	PL-L	NL	PL-NL	EW	VW
$N = 30$	13.82	**13.81**	13.87	13.87	18.72	19.13
$N = 50$	13.86	**13.85**	13.91	13.90	18.86	19.04
$N = 100$	13.49	**13.47**	13.52	13.51	19.51	19.20
$N = 225$	12.77	12.76	12.82	**12.75***	19.87	19.32
$N = 500$	12.14	12.13	12.24	**12.10***	20.62	19.52

All measures are based on 5313 daily out-of-sample returns from December 18, 2000 until January 31, 2022. The best estimator with the lowest SD is indicated in **bold** font. In the columns PL-L, respectively PL-NL, significance at the 0.01 level of outperformance in SD over L, respectively NL, is denoted by an asterisk (*).

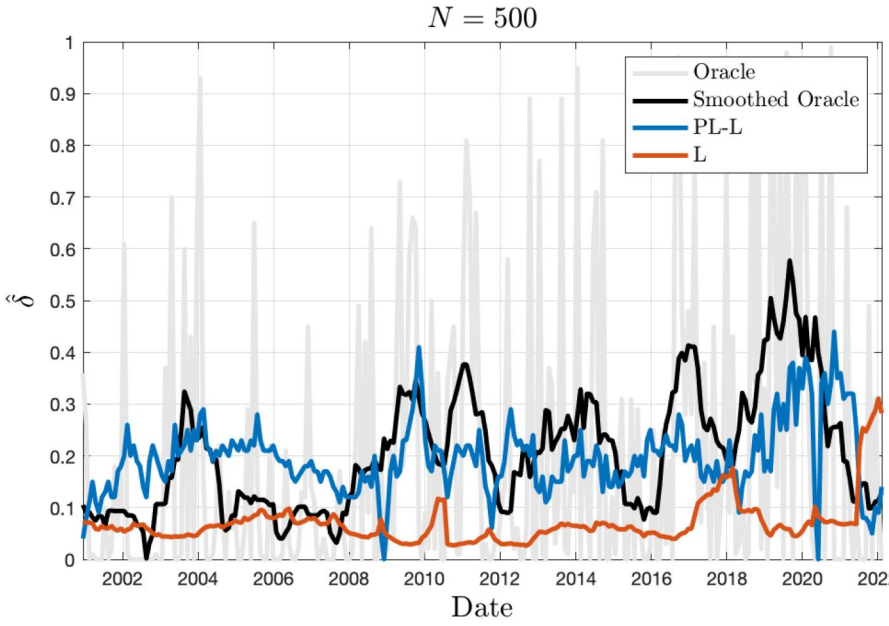


Figure 8. Comparison of the monthly Oracle, PL-L, and L estimated linear shrinkage intensities over time for the $N = 500$ long-only GMV portfolio. For the smoothed Oracle, we use a 1-year moving average smoothing.

The outperformance of PL-L over L is again driven by an increased average shrinkage intensity; see Figure 8. For the $N = 500$ long-only GMV portfolio, the average shrinkage intensity of PL-L is 0.2, which is very close to the Oracle's average shrinkage intensity of 0.21, especially in comparison to L, which has an average shrinkage intensity of 0.07.

However, the outperformance of PL-L is smaller in the long-only setting due to the upper bounds imposed on portfolio weights, which act as constraints in the optimization. As a result, less shrinkage is required due to the implicit shrinkage effect described in Jagannathan and Ma (2003). This reduced need for shrinkage is evident in the lower Oracle and PL-L shrinkage intensities compared to the unconstrained long-short setting, where shrinkage intensity is approximately halved; see Figure 3 for comparison.

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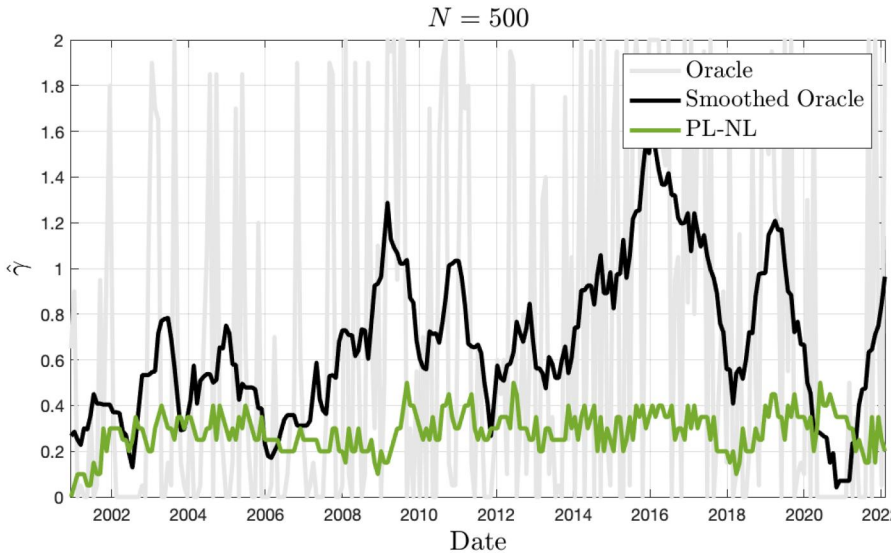


Figure 9. Comparison of the Oracle and PL-NL estimated nonlinear shrinkage intensities adjustment over time for the $N = 500$ long-only GMV portfolio. For the smoothed Oracle, we use a 1-year moving average smoothing.

The reduced need for shrinkage due to the long-only constraint can also be observed for nonlinear shrinkage; see Figure 9. While in the unconstrained setting, a higher shrinkage intensity was required, with an average Oracle adjustment of 1.26, the long-only case requires significantly less shrinkage, with an average Oracle adjustment of 0.68.

PL offers an advantage over classical shrinkage methods by dynamically adjusting shrinkage intensity based on the specific optimization problem and performance metric of interest. Unlike classical approaches, which derive a theoretical optimal shrinkage intensity under the i.i.d. assumption independent of the portfolio optimization step, PL learns to optimize shrinkage in a forward-looking manner, directly incorporating the interaction between shrinkage, constraints and portfolio performance. This adaptive approach allows PL to tailor shrinkage intensity to different portfolio constraints, asset universes, and risk-return objectives, ultimately leading to more robust and economically meaningful improvements in out-of-sample performance.

2.6 Future Research

In this study, we focus on the added value of PL for the computation of shrinkage intensity (ies) without making any assumptions about the data-generating process. However, PL-L and PL-NL remain static covariance matrix estimators. It would therefore be of interest to combine our approach with the dynamic conditional correlation (DCC) model of Engle, Ledoit, and Wolf (2019), specifically integrating the DCC multivariate GARCH model with PL-L or PL-NL for correlation targeting.

Another potential enhancement to improve performance is incorporating higher-frequency data (see, e.g., Callot, Kock, and Medeiros 2017; De Nard et al. 2022) or factor data (see, e.g., De Nard, Ledoit, and Wolf 2021; De Nard and Zhao 2022, 2023; Alves et al. 2024; Fan et al. 2024). An alternative approach is to apply PL not to shrinkage estimators but to other dimensionality reduction techniques such as principal component analysis (see, e.g., Fan, Liao, and Mincheva 2013, 2024).

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Finally, PL could be a powerful tool for learning the optimal time-varying combination of different covariance matrix estimators. For example, PL could be used to determine the shrinkage intensities for multi-target shrinkage estimation (see, e.g., Lancewicki and Aladjem 2014), as well as convex combinations of inverse linear shrinkage estimators (see, e.g., the universal portfolio shrinkage method of Kelly et al. 2025) or, more generally, to derive the optimal weighting scheme for ensemble methods that aggregate multiple covariance matrix estimators.

3 Conclusion

The main contribution of this study is to introduce a data-driven approach to learn the optimal shrinkage intensity(ies) using a supervised policy learning approach. Hence, our findings are grounded in empirical evidence with a practical focus. The proposed PL shrinkage estimator applies to both linear and nonlinear shrinkage, demonstrating improved performance compared to classical shrinkage estimators. The improvement stems from dropping the i.i.d. assumption when computing the shrinkage intensity(ies) and instead leveraging a data-driven approach that does not impose any assumptions on the data-generating process. Notably, the PL shrinkage estimators rely solely on stock return data but can be easily extended to incorporate factor or alternative data signals.

Our results show that an PL agent recognizes that the shrinkage intensity(ies) derived under the i.i.d. assumption are biased downward and automatically adjusts for this bias by increasing shrinkage depending on the market environment. The learned shrinkage intensity (ies) remain stable, intuitive, and correlated with the classical i.i.d. approach. To assess the quality of our covariance matrix estimators, we conduct an extensive empirical study in a high-dimensional setting, focusing on the out-of-sample standard deviation of GMV portfolios, where the PL agent optimizes shrinkage under a minimum-variance utility function.

A further contribution of this study is the application of PL to constrained portfolio optimization, recognizing its practical relevance and the flexibility of PL to adapt to specific investor preferences. Unlike classical shrinkage methods, which are independent of the optimization step, PL dynamically adjusts shrinkage based on the specific constraints, risk-return objectives, and utility functions of the investor. Most classical shrinkage methods tend to shrink too aggressively in constrained optimization problems, as they do not account for the implicit shrinkage impact of constraints, potentially leading to suboptimal portfolio allocations. This adaptability allows PL to incorporate the interaction between shrinkage and portfolio performance in a forward-looking manner, making it particularly well-suited for real-world investment settings where constraints play a crucial role.

Finally, the PL shrinkage estimators enable more efficient implementation of risk-optimized portfolios and are well-suited for real-world investment applications. However, our approach is not limited to portfolio selection or even financial applications; it provides a general framework for improving covariance matrix estimation across various domains.

Funding

None declared.

Conflict of Interest

None declared.

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Appendix A: Proof of Equivalence for Shrinkage toward the Identity Matrix

By substituting the definition of λ_i^{I} from Equation (1.7) into Equation (1.16) and carrying out some algebraic manipulations, we obtain:

$$\tilde{\lambda}_i^{\text{PL-I}} := \widehat{\gamma}\lambda_i^{\mathrm{I}} + (1-\widehat{\gamma})\lambda_i \tag{A1}$$

$$= \widehat{\gamma}(\widehat{\delta}^{\mathrm{I}}\bar{\lambda} + (1-\widehat{\delta}^{\mathrm{I}})\lambda_i) + (1-\widehat{\gamma})\lambda_i \tag{A2}$$

$$= \widehat{\gamma}\widehat{\delta}^{\mathrm{I}}\bar{\lambda} + \widehat{\gamma}(1-\widehat{\delta}^{\mathrm{I}})\lambda_i + (1-\widehat{\gamma})\lambda_i \tag{A3}$$

$$= \widehat{\gamma}\widehat{\delta}^{\mathrm{I}}\bar{\lambda} + (\widehat{\gamma}(1-\widehat{\delta}^{\mathrm{I}}) + 1 - \widehat{\gamma})\lambda_i \tag{A4}$$

$$= \widehat{\gamma}\widehat{\delta}^{\mathrm{I}}\bar{\lambda} + (1-\widehat{\gamma}\widehat{\delta}^{\mathrm{I}})\lambda_i \ . \tag{A5}$$

As $\widehat{\gamma}\widehat{\delta}^{\mathrm{I}} \geq 0$, and for $\widehat{\gamma} \leq 1/\widehat{\delta}^{\mathrm{I}}$, we can write

$$\lambda_i^{\text{PL-I}} = \widehat{\delta}^{\text{PL-I}}\bar{\lambda} + (1-\widehat{\delta}^{\text{PL-I}})\lambda_i \ , \tag{A6}$$

where $\widehat{\delta}^{\text{PL-I}} := \widehat{\gamma}\widehat{\delta}^{\mathrm{I}}$ is the same as in the direct approach (Equation 1.14).

Appendix B: Additional Figures

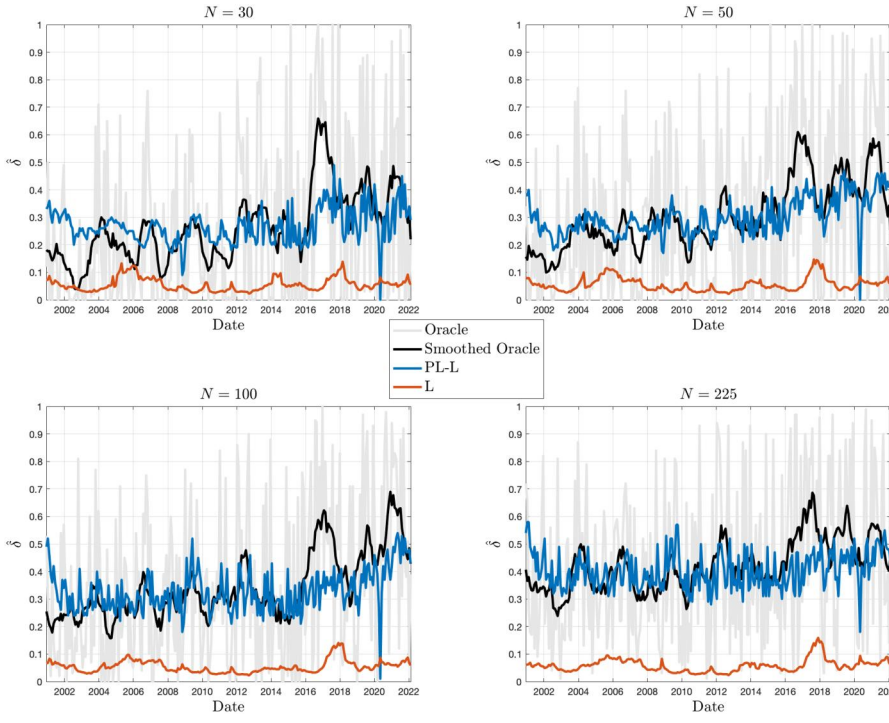


Figure B.1. Comparison of the monthly Oracle, PL-L, and L estimated linear shrinkage intensities over time for various portfolio sizes. For the smoothed Oracle, we use a 1-year moving average smoothing.

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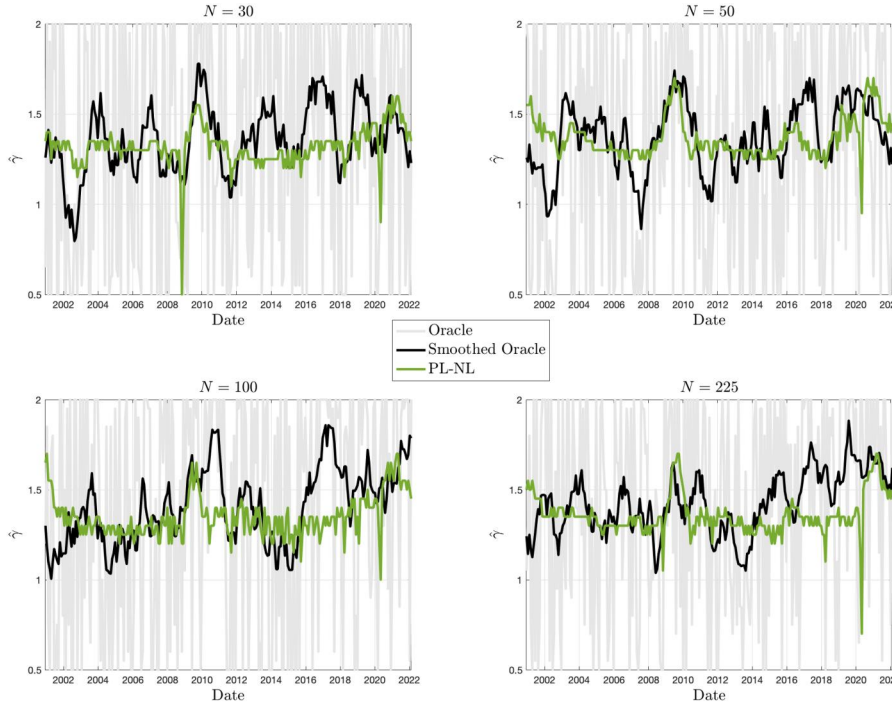


Figure B.2. Comparison of the Oracle and PL-NL estimated nonlinear shrinkage intensities adjustment over time for various portfolio sizes. For the smoothed Oracle, we use a 1-year moving average smoothing.

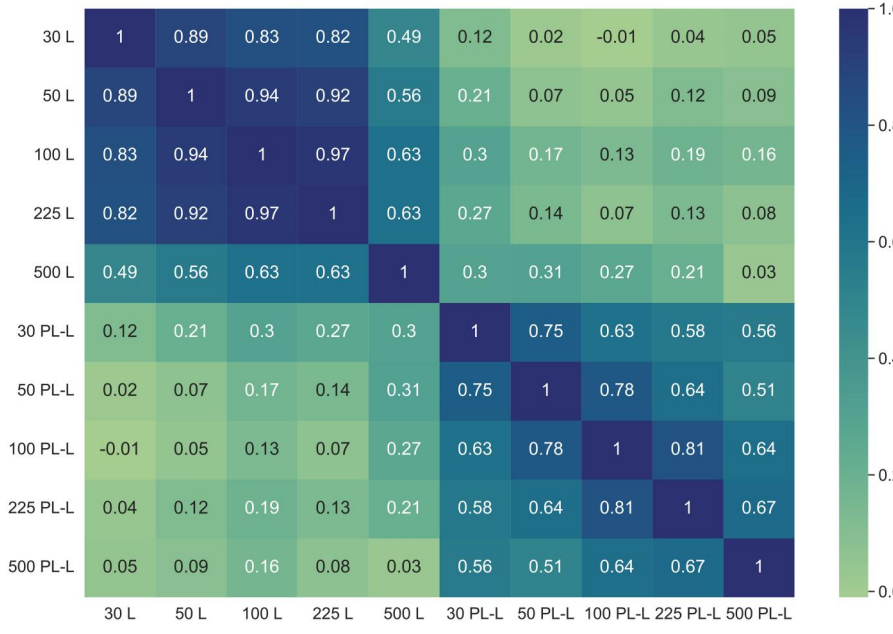


Figure B.3. Correlation matrix of L and PL monthly shrinkage intensities across GMV portfolio sizes. All numbers are based on 253 monthly shrinkage intensities from December 18, 2000 until January 31, 2022.

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